1. introduction

a) Project Idea

- Traffic congestion is a major problem in modern cities.
- It causes long waiting times, delays, and increased fuel consumption.
- **Fixed traffic light timings are not efficient because traffic changes over time.**
- Static systems cannot adapt to real-time traffic conditions.
- This leads to unnecessary congestion and poor traffic flow.
- **Optimization techniques are needed to improve traffic signal control.**
- In this project, AI algorithms **(PSO and GA)** are used to optimize traffic lights.
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Goal of the Project

- Optimize traffic signal timings
- Reduce congestion and waiting time
- Improve overall traffic flow using PSO and GA

b) Problem Formulation & Traffic Modeling

- The system models a traffic network of multiple intersections.
- Each intersection has traffic lights controlling vehicle flow.
- Each traffic light has green signal timing as a decision variable.

Decision Variables

- **Each solution represents traffic light timings:**
  - Example: [NS_green, EW_green, ...]
- These values define how long each direction stays green.
  - **Constraints:**
  - Minimum green time **= 5 seconds**
  - Maximum green time **= 60 seconds**
  -

Objective of the System

- Minimize traffic congestion
- **Reduce** average **waiting time** and **queue length** and **number of stops**

Objective Function

- The system evaluates each solution using:
  - **J = Waiting Time + Queue Length + Number of Stops**
- Lower value = better traffic performance

TRAFFIC_MODES = {
"low": 40, # slow spawning **"medium": 15**, # your current behavior **"high": 5** # fast spawning }

2. Particle Swarm Optimization (PSO)

1. Global Best PSO (gbest PSO)

- All particles share one best solution (global best).
- **Every particle is attracted toward the best solution in the entire swarm.**
- It is inspired by the social behavior of birds and fish.
- It solves problems by improving a group of candidate solutions (particles) over time.

2. PSO Algorithm Steps

- Initialize random particles (traffic signal timings)
- Run traffic simulation for each particle
- Compute fitness score
- Update:
  - Personal Best (pbest)
  - Global Best (gbest)
- Update velocity and position
- Repeat until convergence

3. Velocity Update Rule

$$v = wv + c1r1(pbest - x) + c2r2(gbest - x)$$

- **Inertia (w)** → keeps previous motion
- **Cognitive (c1)** → personal experience = **c1 * r1 * (self.best_pos - self.position)**
- **Social (c2)** → swarm knowledge **= c2 * r2 * (global_best_pos - self.position)**
- **r1, r2** → randomness

4. Parameters Used

- Swarm size (particles) = 6 and 12
- Iterations = 8 and 15
- Inertia weight (w) = 0.7 to balances both behaviors
  - Exploration vs Exploitation
  - Exploration → searching new solutions
  - Exploitation → refining best solutions
- Cognitive coefficient (c1) = 1.5
- Social coefficient (c2) = 1.5
- Velocity limit = [-10, 10] →This controls how much a particle can move in one step.
- Bounds = [5, 60] seconds (green time limits)

PSO SMALL vs PSO BIG - History Table

c1 = 3 c2 = 1

Step	PSO SMALL Best	PSO SMALL Global	PSO SMALL Avg	PSO BIG Best	PSO BIG Global	PSO BIG Avg
0	56.53	56.53	63.30	55.31	55.31	64.26
1	57.05	56.53	63.22	55.43	55.31	60.91
2	57.20	56.53	58.84	53.13	53.13	58.45
3	56.94	56.53	61.35	52.75	52.75	57.15
4	58.27	56.53	61.94	54.25	52.75	55.99
5	58.12	56.53	60.35	52.35	52.35	55.71
6	57.01	56.53	59.35	52.58	52.35	55.75
7	55.40	55.40	58.19	52.78	52.35	56.26
8	-	-	-	52.17	52.17	54.88
9	-	-	-	51.92	51.92	55.13
10	-	-	-	52.04	51.92	55.05
11	-	-	-	52.19	51.92	54.70
12	-	-	-	52.17	51.92	54.20
13	-	-	-	52.19	51.92	54.40
14	-	-	-	52.34	51.92	54.78

PSO SMALL vs PSO BIG - History Table

c1 = 1 c2 = 3

Step	PSO SMALL Best	PSO SMALL Global	PSO SMALL Avg	PSO BIG Best	PSO BIG Global	PSO BIG Avg
0	56.53	56.53	63.30	55.31	55.31	64.26
1	57.05	56.53	63.87	55.43	55.31	59.46
2	55.12	55.12	58.24	53.83	53.83	56.97
3	53.01	53.01	57.12	54.38	53.83	56.64
4	51.90	51.90	55.26	53.66	53.66	55.46
5	50.95	50.95	51.98	52.73	52.73	56.47
6	51.64	50.95	52.68	52.05	52.05	55.02
7	52.21	50.95	52.97	51.98	51.98	54.47
8	-	-	-	52.24	51.98	54.34
9	-	-	-	51.96	51.96	54.49
10	-	-	-	52.28	51.96	55.86
11	-	-	-	51.97	51.96	53.96
12	-	-	-	52.63	51.96	53.79
13	-	-	-	52.17	51.96	53.99
14	-	-	-	51.90	51.90	54.16

PSO SMALL vs PSO BIG - History Table

c1 = 1.5 c2 = 1.5

Step	PSO SMALL Best	PSO SMALL Global	PSO SMALL Avg	PSO BIG Best	PSO BIG Global	PSO BIG Avg
0	56.53	56.53	63.30	55.31	55.31	64.26
1	57.05	56.53	63.69	55.43	55.31	60.81
2	56.21	56.21	58.20	52.95	52.95	56.79
3	54.66	54.66	58.12	53.13	52.95	55.94
4	53.35	53.35	56.69	53.79	52.95	56.15
5	50.68	50.68	54.81	53.07	52.95	55.83
6	51.75	50.68	54.76	53.00	52.95	55.30
7	52.47	50.68	54.70	53.18	52.95	54.59
8	-	-	-	52.95	52.95	55.21
9	-	-	-	52.86	52.86	55.13
10	-	-	-	52.15	52.15	55.06
11	-	-	-	52.15	52.15	54.59
12	-	-	-	51.88	51.88	54.52
13	-	-	-	52.49	51.88	54.38
14	-	-	-	51.90	51.88	54.94

5. Scenario Testing

- PSO was tested under:
  - Low traffic conditions
  - Medium traffic conditions
  - High traffic conditions
- Fitness = evaluate(low traffic) + evaluate(medium traffic) + evaluate(high traffic)
- average of scenarios: Final Fitness = (F_low + F_medium + F_high) / 3

5. Output of PSO

"method": "PSO_SMALL",
"timings": [52,44,6,43],
"score": 50.67588029997983,

--- Iteration 1/8 ---
P01: [40, 6, 20, 17] score=65.3208
P02: [28, 7, 17, 33] score=64.8080
P03: [17, 37, 50, 5] score=69.5547
P04: [58, 24, 10, 10] score=56.6980
P05: [34, 59, 26, 35] score=56.5345
P06: [44, 8, 18, 21] score=66.8855
— iter best=56.5345 global best=56.5345 avg=63.3002

--- Iteration 2/8 ---
P01: [35, 16, 30, 27] score=64.1732
P02: [28, 17, 22, 34] score=70.7616
P03: [27, 47, 40, 15] score=61.8652
P04: [48, 34, 20, 20] score=57.0493
P05: [37, 59, 28, 36] score=60.1501
P06: [37, 18, 18, 25] score=68.1576
— iter best=57.0493 global best=56.5345 avg=63.6928

--- Iteration 3/8 ---
P01: [31, 26, 33, 37] score=60.0386
P02: [30, 27, 24, 35] score=58.9599
P03: [35, 56, 30, 25] score=58.0137
P04: [43, 41, 23, 25] score=56.6066
P05: [32, 58, 23, 35] score=59.3771
P06: [37, 28, 20, 28] score=56.2123
— iter best=56.2123 global best=56.2123 avg=58.2014

"method": "PSO_BIG",
"timings": [50,25,45,34],
"score": 51.880999509600436,

--- Iteration 1/15 ---
P01: [40, 6, 20, 17] score=65.9509
P02: [28, 7, 17, 33] score=64.2071
P03: [17, 37, 50, 5] score=73.8641
P04: [58, 24, 10, 10] score=61.0108
P05: [34, 59, 26, 35] score=58.2967
P06: [44, 8, 18, 21] score=67.4338
P07: [40, 25, 25, 17] score=59.3944
P08: [14, 45, 14, 26] score=71.8012
P09: [51, 48, 18, 7] score=56.4218
P10: [53, 22, 41, 27] score=55.3144
P11: [36, 19, 37, 54] score=58.2478
P12: [10, 8, 11, 40] score=79.1704
— iter best=55.3144 global best=55.3144 avg=64.2595

--- Iteration 2/15 ---
P01: [50, 16, 30, 26] score=58.6800
P02: [34, 10, 27, 31] score=67.6402
P03: [27, 34, 46, 9] score=63.8278
P04: [60, 24, 15, 13] score=60.0191
P05: [44, 49, 36, 26] score=56.7803
P06: [44, 11, 23, 22] score=66.4575
P07: [41, 27, 30, 20] score=57.3415
P08: [24, 35, 24, 28] score=63.0702
P09: [51, 38, 28, 17] score=57.9722
P10: [56, 22, 39, 25] score=55.4341
P11: [40, 18, 42, 47] score=57.6224
P12: [20, 18, 21, 30] score=64.8358
— iter best=55.4341 global best=55.3144 avg=60.8068

3. Genetic Algorithm (GA)

1. Definition

- Genetic Algorithm (GA) is a population-based optimization technique.
- It is inspired by the process of natural evolution.
- It improves solutions over generations using **selection**, **crossover**, and **mutation**.

2. GA Algorithm Steps

- Initialize random population of solutions
- Evaluate fitness of each individual using simulation
- Select best individuals (tournament selection)
- Apply crossover (combine parents)
- Apply mutation (random changes)
- Create new generation
- Repeat for multiple generations

3. Genetic Operators

Selection **(Tournament Selection)**

- Randomly pick 3 individuals
- Choose the best among them

Crossover **(Single-Point Crossover)**

- Combines two parents to create offspring

Mutation(**Random Reset** / Additive Mutation (step-based mutation))

- Randomly modifies genes

Elitism

- Best solution is always preserved

4. Parameters Used

- Population size = 6–12 individuals
- Generations = 8–15
- Crossover rate = 0.9 (chance that two parents will perform crossover)
- Mutation rate = 0.3 (chance of being mutated)
- Tournament size = 3
- Elite size = 1
- Bounds = 5–60 seconds (green time limits)

GA SMALL vs GA BIG - History Table

c = 0.9 m= 0.3

Step	GA SMALL Best	GA SMALL Global	GA SMALL Avg	GA BIG Best	GA BIG Global	GA BIG Avg
0	58.06	58.06	67.32	52.23	52.23	64.86
1	53.36	53.36	57.78	52.23	52.23	58.45
2	51.29	51.29	54.41	52.23	52.23	56.94
3	51.16	51.16	53.45	52.23	52.23	56.04
4	50.79	50.79	52.07	50.41	50.41	54.58
5	49.31	49.31	50.91	50.41	50.41	54.48
6	49.18	49.18	50.83	50.41	50.41	53.31
7	49.18	49.18	49.93	50.41	50.41	52.07
8	-	-	-	50.41	50.41	53.68
9	-	-	-	50.41	50.41	52.67
10	-	-	-	50.30	50.30	53.00
11	-	-	-	50.30	50.30	51.68
12	-	-	-	50.30	50.30	52.49
13	-	-	-	50.30	50.30	53.62
14	-	-	-	50.30	50.30	53.32

GA SMALL vs GA BIG - History Table

c = 0.3 m= 0.9

Step	GA SMALL Best	GA SMALL Global	GA SMALL Avg	GA BIG Best	GA BIG Global	GA BIG Avg
0	58.06	58.06	67.32	52.23	52.23	64.86
1	52.97	52.97	60.15	50.37	50.37	58.73
2	50.00	50.00	54.19	50.37	50.37	56.27
3	49.92	49.92	51.40	50.37	50.37	56.72
4	49.92	49.92	52.36	50.37	50.37	58.38
5	49.92	49.92	51.83	50.37	50.37	56.23
6	49.77	49.77	51.39	50.37	50.37	57.47
7	49.77	49.77	53.10	50.30	50.30	56.19
8	-	-	-	50.30	50.30	55.69
9	-	-	-	50.30	50.30	55.52
10	-	-	-	50.30	50.30	56.59
11	-	-	-	50.30	50.30	56.39
12	-	-	-	50.30	50.30	55.52
13	-	-	-	50.30	50.30	55.94
14	-	-	-	50.30	50.30	55.59

GA SMALL vs GA BIG - History Table

c = 0.5 m= 0.5

Step	GA SMALL Best	GA SMALL Global	GA SMALL Avg	GA BIG Best	GA BIG Global	GA BIG Avg
0	58.06	58.06	67.32	52.23	52.23	64.86
1	54.03	54.03	58.80	52.23	52.23	58.24
2	51.16	51.16	53.20	51.35	51.35	55.65
3	50.80	50.80	55.08	51.35	51.35	56.35
4	50.25	50.25	53.28	50.22	50.22	54.18
5	50.25	50.25	52.22	50.22	50.22	54.40
6	50.25	50.25	54.28	50.22	50.22	54.40
7	49.41	49.41	51.08	50.22	50.22	53.73
8	-	-	-	50.22	50.22	54.38
9	-	-	-	50.20	50.20	54.43
10	-	-	-	50.17	50.17	54.60
11	-	-	-	50.17	50.17	54.07
12	-	-	-	49.97	49.97	54.34
13	-	-	-	49.97	49.97	55.50
14	-	-	-	49.97	49.97	54.50

5. Scenario Testing

- GA was tested under:
    - Low traffic conditions
    - Medium traffic conditions
    - High traffic conditions
- Fitness = evaluate(low traffic) + evaluate(medium traffic) + evaluate(high traffic)
- average of scenarios: Final Fitness = (F_low + F_medium + F_high) / 3

5. Output of GA

"method": "GA_SMALL",
"timings": [60,5,43,20],
"score": 49.41093148052207,

--- Generation 1/8 ---
01: [45, 12, 6, 52] → 61.6073
02: [22, 20, 19, 13] → 69.2275
03: [52, 11, 48, 52] → 58.0592
04: [39, 10, 42, 32] → 63.4744
05: [7, 6, 10, 18] → 86.7596
06: [19, 37, 43, 6] → 64.7959
— gen best=58.0592 global best=58.0592 avg=67.3206

--- Generation 2/8 ---
01: [52, 11, 48, 52] → 58.0592
02: [35, 10, 5, 56] → 66.8179
03: [53, 11, 54, 46] → 55.8534
04: [45, 12, 6, 52] → 61.6073
05: [60, 11, 46, 47] → 56.4477
06: [51, 14, 6, 49] → 54.0340
— gen best=54.0340 global best=54.0340 avg=58.8032

--- Generation 3/8 ---
01: [51, 14, 6, 49] → 54.0340
02: [53, 7, 54, 46] → 56.7139
03: [51, 20, 5, 47] → 53.0485
04: [51, 14, 46, 38] → 51.1647
05: [60, 11, 13, 44] → 52.0234
06: [60, 14, 6, 57] → 52.1887
— gen best=51.1647 global best=51.1647 avg=53.1955

"method": "GA_BIG",
"timings": [60,5,54,39],
"score": 49.97200830071484,

--- Generation 1/15 ---
01: [45, 12, 6, 52] → 55.7945
02: [22, 20, 19, 13] → 75.6559
03: [52, 11, 48, 52] → 58.5167
04: [39, 10, 42, 32] → 59.5545
05: [7, 6, 10, 18] → 92.2915
06: [19, 37, 43, 6] → 67.9091
07: [40, 17, 50, 46] → 60.3949
08: [49, 39, 31, 19] → 57.0316
09: [33, 42, 22, 56] → 59.5676
10: [60, 5, 53, 56] → 52.2292
11: [15, 49, 32, 26] → 65.6327
12: [22, 14, 18, 53] → 73.6954
— gen best=52.2292 global best=52.2292 avg=64.8561

--- Generation 2/15 ---
01: [60, 5, 53, 56] → 52.2292
02: [35, 10, 5, 36] → 61.8284
03: [40, 10, 48, 46] → 58.5764
04: [40, 17, 50, 46] → 60.3949
05: [25, 49, 30, 21] → 64.1719
06: [45, 12, 42, 29] → 59.8475
07: [58, 40, 25, 15] → 61.4700
08: [60, 5, 53, 56] → 52.2292
09: [49, 45, 22, 17] → 60.9882
10: [52, 20, 53, 47] → 56.1494
11: [60, 5, 55, 47] → 52.7048
12: [52, 14, 48, 52] → 58.3274
— gen best=52.2292 global best=52.2292 avg=58.2431

4. Objective of Comparison

- Both PSO and GA were applied to optimize traffic signal timings.
- Both algorithms were tested under identical traffic scenarios.
- The goal is to **compare performance** in terms of:
  - fitness score
  - convergence behavior
  - traffic efficiency metrics

Metric	PSO_SMALL	GA_SMALL
Best Fitness Score	**50.67**	**49.41**
Best Timings	[52, 44, 6, 43]	[60, 5, 43, 20]

1. Observation

- GA achieved a slightly better final fitness score (49.41 < 50.67)
- GA found a more optimal solution at the end
- PSO solution is close but slightly less optimal

2. Performance Across Traffic Scenarios

Low Traffic

Algorithm	Fitness
PSO	48.66
GA	48.15

Medium Traffic

Algorithm	Fitness
PSO	55.78
GA	52.18

High Traffic

Algorithm	Fitness
PSO	62.04
GA	53.03

3. PSO vs GA Performance Comparison Table

1. Low Traffic Scenario

Metric	PSO	GA	Better
Fitness	48.66	**48.15**	GA
Avg Waiting Time	19.73	**16.79**	GA
Queue Length	15	**16 (slightly worse)**	PSO
Stops	15	**17**	PSO

2. Medium Traffic Scenario

Metric	PSO	GA	Better
Fitness	55.78	**52.18**	GA
Avg Waiting Time	16.5	**14.72**	GA
Queue Length	49	**41**	GA
Stops	16	**19**	PSO (slightly)

3. High Traffic Scenario

Metric	PSO	GA	Better
Fitness	62.04	**53.03**	GA
Avg Waiting Time	16.46	**13.03**	GA
Queue Length	71	**67**	GA
Stops	23	**20**	GA

Overall Average Comparison

Metric	PSO (Avg)	GA (Avg)	Winner
Fitness	Higher	**Lower**	GA
Waiting Time	Higher	**Lower**	GA
Queue Length	Higher	**Lower**	GA

4. Key Comparison Insights

PSO Strengths

- **Faster convergence**
- **Good early optimization**
- Stable behavior

PSO Weakness

- Gets stuck earlier
- Less effective in high traffic complexity

GA Strengths

- **Better final solution**
- Strong performance in all traffic levels
- Better exploration of solution space

GA Weakness

- **Slower convergence**
- More fluctuations during early generations

PSO is faster but less optimal

Reason 1: Strong social learning

- PSO particles **follow global best quickly**
- So convergence is fast

That is why PSO reaches good solution early (50.67 at iteration 5–6)

Reason 2: Less diversity

- All particles **move toward same best** solution
- Diversity decreases over time

Result:

- **faster convergence**
- **but limited exploration**

so GA is better

Reason 1: Better exploration

- GA uses:
  - crossover (**mixing solutions**)
  - mutation (**random changes**)

This allows GA to explore more diverse traffic signal combinations.

Reason 2: Avoiding early convergence

- PSO converges quickly toward global best
- But this causes:
  - **reduced exploration**

Reason 3: Traffic problem is highly non-linear
traffic simulation includes:

- queues
- congestion spikes
- vehicle interactions

This creates a complex search space

GA handles this better because:

- **randomness + mutation helps explore** irregular solutions
- **PSO is more "directional" and can miss hidden better solutions**